

ABC MANUFACTURING CORPORATION

Environmental Health & Safety Department

STANDARD OPERATING PROCEDURE

Chemical Handling and Storage

Document No. SOP-EHS-001

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A. Document Control

A.1 Revision History

Rev	Date	Description of Change	Author	Approver
00	12 Feb 2019	Initial issue.	J. Alvarez	M. Okafor
01	20 Mar 2021	Added secondary containment requirements (NFPA 30) and updated PPE matrix.	J. Alvarez	M. Okafor
02	05 Aug 2023	Aligned classification language to GHS/HazCom 2012 update; added incompatible-materials segregation chart.	R. Nguyen	M. Okafor
03	15 Jan 2026	Biennial review. Added risk matrix, revised emergency notification chain, added receiving-inspection checklist (Appendix A).	J. Alvarez	M. Okafor

A.2 Distribution List

Copy No.	Holder / Role	Location	Format
Master	Document Owner — Manager, EHS	DMS (electronic master)	Electronic
01	Production Floor Supervisors	Line 1–4 supervisor stations	Controlled print
02	Chemical Store / Receiving Lead	Chemical storage building	Controlled print
03	Maintenance Planner	Maintenance office	Electronic
04	Training Coordinator	Learning Management System	Electronic
05	Emergency Response Team Lead	ERT muster point	Controlled print

A.3 Controlled Copy Statement

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1. Purpose

This Standard Operating Procedure (SOP) establishes the minimum requirements and controls for the safe receipt, handling, movement, storage, and inventory of hazardous chemicals at ABC Manufacturing Corporation. Its purpose is to protect employees, contractors, visitors, property, and the environment from the hazards associated with hazardous chemicals and to ensure compliance with applicable regulatory and internal management-system requirements.

- Prevent uncontrolled release, fire, reaction, or exposure arising from chemical handling and storage.
- Ensure chemicals are segregated by compatibility and stored within their design conditions.
- Define roles, training, personal protective equipment (PPE), and documentation required to perform chemical handling safely.
- Support the objectives of the site ISO 45001 occupational health and safety and ISO 14001 environmental management systems.

2. Scope

This procedure applies to all ABC Manufacturing personnel, contractors, and temporary workers who receive, transport, dispense, use, or store hazardous chemicals at any ABC Manufacturing facility, including the chemical storage building, point-of-use satellite storage, production lines, laboratories, and maintenance areas.

2.1 Inclusions

- Flammable and combustible liquids, corrosives, oxidizers, toxics, and water-reactive materials in containers up to and including intermediate bulk containers (IBCs).
- Receiving inspection, container labeling, secondary containment, and inventory reconciliation.

2.2 Exclusions

- Bulk storage tanks and piped chemical distribution systems (governed by SOP-EHS-004, Chemical Transfer Procedure, and the site Process Safety program).
- Compressed gases in cylinders (governed by SOP-EHS-005, Compressed Gas Cylinder Safety).
- Chemical spill response actions (governed by SOP-EHS-003, Chemical Spill Response). This SOP addresses spill prevention and initial containment only.

3. References

This procedure is based on publicly available regulations, consensus standards, and management-system requirements. Where a conflict exists, the most stringent applicable requirement governs.

Reference	Title / Subject
29 CFR 1910.106	Flammable liquids
29 CFR 1910.1200	Hazard Communication (HazCom / GHS)
29 CFR 1910.120	Hazardous waste operations and emergency response (HAZWOPER)
29 CFR 1910.132-.138	Personal protective equipment (general, eye/face, hand)
29 CFR 1910.157	Portable fire extinguishers
NFPA 30	Flammable and Combustible Liquids Code
NFPA 704	Standard System for the Identification of the Hazards of Materials for Emergency Response
ANSI/ISEA Z358.1	Emergency Eyewash and Shower Equipment
ISO 45001:2018	Occupational health and safety management systems

Reference	Title / Subject
ISO 14001:2015	Environmental management systems
ISO 9001:2015	Quality management systems (document control)
Internal	SOP-EHS-002 HazCom; SOP-EHS-003 Spill Response; SOP-EHS-004 Chemical Transfer; SOP-EHS-005 Compressed Gas

4. Definitions

Term	Definition
Hazardous Chemical	Any chemical classified as a physical or health hazard under the Hazard Communication Standard, or otherwise capable of causing harm to persons, property, or the environment.
SDS	Safety Data Sheet — the standardized 16-section document supplied by the manufacturer describing a chemical's hazards, handling, and emergency measures.
Secondary Containment	A system such as a spill pallet, curbed area, or dike designed to capture leaks or spills from primary containers before they reach personnel, drains, or the environment.
Incompatible Materials	Chemicals that, when mixed, may react to produce heat, fire, toxic gas, or pressure (e.g., acids with bases, oxidizers with flammables).
Satellite Storage	A small quantity of chemical stored at or near the point of use, within limits defined by NFPA 30 and this SOP.
Competent Person	An individual who, by training and experience, is capable of identifying chemical hazards and is authorized to take corrective action.
NFPA 704 Diamond	The four-color placard indicating health (blue), flammability (red), instability (yellow), and special (white) hazards on a 0–4 scale.

5. Responsibilities

Role	Responsibility
EHS Manager (Document Owner)	Maintains this SOP, approves chemical additions to the approved list, ensures training and audits are conducted.
Area Supervisor	Ensures only trained personnel handle chemicals, verifies PPE availability, confirms storage compliance each shift.
Chemical Store / Receiving Lead	Performs receiving inspection, verifies labels and SDS availability, maintains segregation and inventory records.
Employee / Operator	Follows this SOP, inspects containers before use, wears required PPE, reports damaged containers, leaks, and near-misses.
Maintenance Technician	Maintains ventilation, containment, eyewash/shower, and storage infrastructure per PM schedule.
Emergency Response Team (ERT)	Responds to releases beyond initial containment capacity per SOP-EHS-003.

6. Required Training

Personnel must complete the following training before performing tasks covered by this SOP. Training records are retained in the LMS and are subject to audit.

Training Module	Audience	Frequency
Hazard Communication / GHS label & SDS literacy	All chemical handlers	Initial + annual
Chemical Handling & Storage (this SOP)	All chemical handlers	Initial + on revision
Secondary containment & spill prevention	Chemical store, operators	Initial + 2-yr refresh
Emergency eyewash/shower use	All chemical handlers	Initial + annual
Incompatible-materials segregation	Chemical store / receiving lead	Initial + 2-yr refresh

7. Required Qualifications

- Minimum 18 years of age and authorized by the Area Supervisor to perform chemical handling.
- Current completion of all training modules in Section 6, verified in the LMS.
- Medical clearance to wear required PPE, including respiratory protection where the SDS or task assessment requires it.
- Demonstrated competency during on-the-job evaluation signed off by a Competent Person.

8. Required PPE

PPE selection is task- and chemical-specific and must be confirmed against Section 8 of the applicable SDS before work begins. The following is the minimum baseline.

Task	Eye/Face	Hand	Body	Foot	Respiratory
General handling (sealed containers)	Safety glasses	Nitrile gloves	—	Safety shoes	—
Dispensing / open transfer of corrosives	Goggles + face shield	Chemical-resistant gauntlet	Apron	Safety shoes	Per SDS / assessment
Handling flammables	Safety glasses	Nitrile gloves	FR-rated as required	Safety shoes (static-dissipative)	—
Cleanup of incidental spill (< 1 gal)	Goggles	Chemical-resistant gloves	Coverall	Chemical boots	Per SDS

CAUTION

Glove material must match the chemical. Nitrile is not universal — verify breakthrough time against the SDS. Never assume one glove protects against all chemicals.

9. Required Equipment

- Approved flammable-storage cabinets (self-closing, self-latching, vented per site design) and corrosive/acid cabinets as applicable.
- Secondary containment: spill pallets, drum trays, or curbed storage sufficient to contain 110% of the largest container or 10% of aggregate volume, whichever is greater (NFPA 30 principle).
- Approved container types (bonded/grounded safety cans for flammable dispensing), drum trucks, and non-sparking tools where flammables are present.
- ANSI/ISEA Z358.1-compliant emergency eyewash and drench shower within 10 seconds' travel of the handling area.
- Suitable portable fire extinguisher (e.g., Class B for flammable liquids) and spill kit rated for the stored chemicals.
- NFPA 704 placarding at storage-room entrances and calibrated inventory / label printer.

10. Hazard Identification

The following hazards are inherent to chemical handling and storage and must be controlled through the measures in this SOP.

- **Fire / explosion** — ignition of flammable vapors from spills, static discharge, or incompatible storage.
- **Chemical exposure** — skin/eye contact, inhalation of vapors, or ingestion causing acute or chronic health effects.
- **Reactive hazards** — heat, gas, or pressure generated when incompatible materials contact one another.
- **Environmental release** — spill reaching drains, soil, or waterways.
- **Ergonomic / mechanical** — manual handling of drums and containers; struck-by or crush from moving containers.
- **Corrosion / container failure** — degraded, dented, or over-age containers leaking product.

11. Risk Assessment

Risk level is derived from the severity × likelihood matrix below. Any residual risk assessed as High or Critical requires additional controls and Area Supervisor authorization before work proceeds.

11.1 Risk Rating Matrix

Severity ↓ / Likelihood →	Rare	Unlikely	Possible	Likely	Almost Certain
Catastrophic	Medium	High	High	Critical	Critical
Major	Medium	Medium	High	High	Critical
Moderate	Low	Medium	Medium	High	High
Minor	Low	Low	Medium	Medium	High
Negligible	Low	Low	Low	Low	Medium

11.2 Task Risk Register

Hazard	Severity	Likelihood	Risk Level	Control Measures
Flammable vapor ignition during dispensing	Catastrophic	Unlikely	High	Bonding/grounding, ventilation, no ignition sources, Class B extinguisher, static-dissipative footwear.
Incompatible chemicals stored together	Major	Possible	High	Segregation per compatibility chart (App. C), separate cabinets, receiving verification.
Corrosive splash to eyes/skin	Major	Possible	High	Goggles/face shield, gauntlet gloves, Z358.1 eyewash within 10 s, dispensing at low flow.
Spill reaching floor drain	Moderate	Possible	Medium	Secondary containment (110%), drain covers, spill kit, curbed storage.
Drum handling — musculoskeletal / crush	Moderate	Likely	High	Drum truck/lift aids, two-person lift limits, trained handling, PPE.
Container degradation / leak in storage	Moderate	Unlikely	Medium	FIFO stock rotation, monthly inspection, containment trays, condition tagging.

12. Detailed Step-by-Step Procedure

12.1 Pre-Job Preparation

1. Identify the chemical(s) to be handled and obtain the current SDS. Confirm hazard class, PPE (SDS §8), storage, and incompatibilities (SDS §7, §10).

2. Complete a pre-task hazard review; confirm the task risk rating (Section 11) is Medium or lower, or obtain supervisor authorization for High/Critical.
3. Verify that trained personnel, required PPE, spill kit, fire extinguisher, and eyewash/shower are available and serviceable.
4. Confirm ventilation is operating and ignition sources are controlled in areas where flammables will be handled.

NOTE

If any control listed in the pre-task review is missing or out of service, STOP and notify the Area Supervisor. Do not begin work.

12.2 Equipment & Container Inspection

1. Inspect each container for dents, corrosion, bulging, leaks, and legible GHS labeling. Reject and quarantine any container that fails.
2. Verify secondary containment is clean, intact, and capable of holding 110% of the largest container.
3. Inspect dispensing equipment: bonding/grounding cables and clamps, pump/valve integrity, and safety-can flame arrestor.
4. Confirm storage cabinet self-closes and self-latches; confirm NFPA 704 placarding is correct at the storage entrance.

12.3 Work Execution — Receiving & Storage

1. At receiving, verify the delivery against the purchase order, confirm the SDS is on file, and check the label matches the product.
2. Record the receipt in the chemical inventory (product, quantity, lot, receipt date) and apply a site container/date label for FIFO rotation.
3. Transport containers using approved drum trucks or lift aids; never roll or drag drums. Keep containers upright and closures tight.
4. Place chemicals in the correct storage location by compatibility group (Appendix C). Never store incompatible materials in shared containment.

WARNING

Never store acids with bases, oxidizers with flammables/organics, or water-reactives near water sources. Incompatible storage can cause fire, toxic gas, or violent reaction.

5. Store flammable liquids in approved flammable-storage cabinets; keep quantities at the point of use within satellite-storage limits.
6. Keep aisles clear, containers off the floor where practical on shelving rated for the load, and heavier containers on lower shelves.

12.4 Work Execution — Handling & Dispensing

1. Don PPE per Section 8 and the SDS. Position the eyewash/drench shower within reach before opening any container.
2. For flammable transfer, bond the source and receiving containers and ground the assembly before opening. Dispense with the container over containment.
3. Open containers slowly to relieve pressure; pour or pump at a controlled rate to avoid splashing or vapor generation.
4. Reseal containers immediately after use. Return chemicals to their designated storage; do not leave open containers unattended.

12.5 Verification

1. Confirm all containers are closed, labeled, and returned to compatible storage; confirm bonding/grounding is disconnected and stored.
2. Verify no spills, drips, or residue remain and that secondary containment is dry and clean.
3. Reconcile inventory: update quantities used and confirm running balance matches physical stock.

12.6 Shutdown

1. De-energize and secure dispensing equipment; verify pumps and valves are closed and leak-free.
2. Confirm storage cabinets are latched, the storage room is secured, and ventilation continues per design where required.

12.7 Housekeeping

1. Wipe down work surfaces and dispensing equipment; dispose of contaminated materials as hazardous waste per site procedure.
2. Restock the spill kit and PPE stations; report any consumables used to the Area Supervisor.
3. Ensure aisles, exits, eyewash access, and extinguisher access are clear before leaving the area.

12.8 Documentation

1. Complete the applicable records in Section 14 (receiving inspection, storage inspection, inventory update).
2. File records in the DMS; retain per the site records-retention schedule and ISO 9001 document control requirements.

12.9 Escalation Procedures

1. Damaged/leaking container within containment capacity → contain, isolate, notify Area Supervisor, tag out, arrange disposal.
2. Spill or exposure beyond initial containment capacity → initiate Emergency Response (Section 13) and SOP-EHS-003.
3. Repeated near-misses or an unsafe condition that cannot be corrected → stop work, notify EHS, log per SOP-EHS-013/014.

13. Emergency Response

WARNING

Do not attempt cleanup or rescue beyond your training and available controls. Protect life first, then contain. When in doubt, evacuate and call for help.

13.1 Immediate Actions

- Alert others in the area, stop the source if safe to do so, and evacuate personnel upwind/uphill of the release.
- Eliminate ignition sources for flammable releases; do not operate switches that could spark.

13.2 Medical Response

- Skin/eye contact: flush at the eyewash/drench shower for at least 15 minutes; remove contaminated clothing; seek medical aid. Provide the SDS to responders.
- Inhalation: move the person to fresh air; if breathing is impaired, call emergency services and begin first aid by trained personnel.

13.3 Fire Response

- For a small, incipient Class B fire and only if trained, use a Class B extinguisher; otherwise activate the alarm and evacuate.
- Never use water on a flammable-liquid or water-reactive fire unless the SDS/pre-plan specifically directs it.

13.4 Spill Response

- Incidental spill within containment and within training: don PPE, contain with the spill kit, and clean up per SOP-EHS-003.
- Release to drain, large volume, or unknown material: evacuate, isolate the area, and summon the Emergency Response Team.

13.5 Equipment Shutdown

- Where safe, isolate energy and close valves/pumps feeding the release; do not re-enter to shut down if conditions are unsafe.

13.6 Notification Chain

Step	Notify	Contact / Method	When
1	Area Supervisor	Radio / floor phone	Immediately on any release or exposure
2	EHS On-Call	Ext. 2400 / site radio	Within 5 minutes
3	Emergency Response Team	Emergency line 2222	Release beyond containment / fire / injury
4	External Emergency Services	911	Life threat, fire, or major release
5	EHS Manager & Plant Manager	Per call tree	Within 30 minutes; recordable/reportable events

14. Required Documentation

Record	Purpose	Owner	Retention
Receiving Inspection Checklist (App. A)	Verify condition, label, SDS at receipt	Receiving Lead	3 years
Storage Area Inspection Checklist (App. B)	Monthly verification of storage compliance	Area Supervisor	3 years
Training Record (LMS)	Evidence of qualified personnel	Training Coordinator	Duration + 3 years
Chemical Inventory / Maintenance Log	Stock control and equipment PM	Chemical Store Lead	3 years
Incident / Near-Miss Report (App. E)	Capture releases, exposures, near-misses	EHS	5 years
Hot area permits (if applicable)	Cross-reference to hot work near flammables	Area Supervisor	Per SOP-EHS-006

15. References (Public Standards)

- OSHA 29 CFR 1910.106 — Flammable liquids.
- OSHA 29 CFR 1910.1200 — Hazard Communication.
- OSHA 29 CFR 1910.120 — HAZWOPER.
- OSHA 29 CFR 1910.132–.138 — Personal protective equipment.
- OSHA 29 CFR 1910.157 — Portable fire extinguishers.
- NFPA 30 — Flammable and Combustible Liquids Code.
- NFPA 704 — Identification of the Hazards of Materials for Emergency Response.
- ANSI/ISEA Z358.1 — Emergency Eyewash and Shower Equipment.
- ISO 45001:2018; ISO 14001:2015; ISO 9001:2015 — Management-system standards.

16. Appendices

Appendix A — Chemical Receiving Inspection Checklist

#	Inspection Item	Pass	Fail	N/A
1	Delivery matches purchase order (product, quantity, lot)	☐	☐	☐
2	Container intact — no dents, corrosion, bulging, or leaks	☐	☐	☐

#	Inspection Item	Pass	Fail	N/A
3	GHS label present, legible, and matches product	☐	☐	☐
4	Current SDS available in the DMS	☐	☐	☐
5	Product is on the site approved chemical list	☐	☐	☐
6	Compatible storage location identified and available	☐	☐	☐
7	Site container/date label applied for FIFO	☐	☐	☐
8	Secondary containment available and serviceable	☐	☐	☐

Inspected by: _____ Date: _____ Disposition (Accept / Quarantine / Reject): _____

Appendix B — Monthly Storage Area Inspection Checklist

#	Item	OK	Action Needed
1	Cabinets self-close and self-latch; vents clear	☐	☐
2	Incompatible materials properly segregated	☐	☐
3	Secondary containment clean, dry, ≥110% capacity	☐	☐
4	NFPA 704 placarding correct and legible	☐	☐
5	Eyewash/shower serviceable (weekly activation logged)	☐	☐
6	Spill kit stocked; extinguisher charged and accessible	☐	☐
7	Aisles/exits clear; no expired stock (FIFO)	☐	☐
8	Ventilation operating where required	☐	☐

Appendix C — Incompatible Materials Segregation Chart

Store separately →	Flammables	Oxidizers	Acids	Bases	Water-Reactives
Flammables	OK	NO	CAUTION	CAUTION	NO
Oxidizers	NO	OK	CAUTION	CAUTION	NO
Acids	CAUTION	CAUTION	OK	NO	NO
Bases	CAUTION	CAUTION	NO	OK	CAUTION
Water-Reactives	NO	NO	NO	CAUTION	OK

Legend: OK = may be co-located within compatible containment • CAUTION = separate distance / containment required • NO = must be physically separated. Always defer to the SDS §7/§10.

Appendix D — Chemical Handling Decision Flow

START → Is the SDS on file and current? — NO → Obtain SDS before proceeding.

YES → Is the container intact and correctly labeled? — NO → Quarantine and report (Section 12.9).

YES → Is trained personnel + required PPE + eyewash/spill kit available? — NO → STOP; notify Area Supervisor.

YES → Is the task risk Medium or lower (Section 11)? — NO → Obtain supervisor authorization / add controls.

YES → Proceed with handling per Section 12 → Verify, document, and return to compatible storage → END.

Appendix E — Incident / Near-Miss Report Form

Field	Entry
Report No.	
Date / Time of event	
Location / Area	
Reported by	
Type (check one)	☐ Injury ☐ Exposure ☐ Spill/Release ☐ Fire ☐ Near-Miss ☐ Property
Chemical(s) involved	
Description of event	
Immediate action taken	
Injuries / medical treatment	
Notifications made (who/when)	
Supervisor signature / date	
EHS follow-up (RCA ref. SOP-EHS-015)	